

AI-POWERED SMART MUNICIPAL ASSET MANAGEMENT & MAINTENANCE PLATFORM

Comprehensive Software Requirement Specification (SRS) Document

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3. Introduction

3.1 Purpose

This Software Requirement Specification (SRS) document details the complete functional and non-functional requirements for the **AI-Powered Smart Municipal Asset Management and Maintenance Platform**. The document establishes a rigorous engineering contract between the municipal corporation governance, engineering teams, system architects, and municipal operational staff. Grounded in the IEEE Std 830-1998 standards, this specification serves as the foundational reference for architectural design, system implementation, test validation, and software procurement.

3.2 Scope

The AI-Powered Smart Municipal Asset Management and Maintenance Platform is an enterprise-grade, cloud-native IoT and AI solution engineered to revolutionize municipal asset lifecycle management. The system provides a unified digital ecosystem covering public infrastructure assets including roads, bridges, streetlights, drainage systems, water supply pipelines, public parks, and civic buildings.

Key capabilities encompass real-time telemetry ingestion via IoT sensors, high-precision GIS spatial mapping, automated degradation modeling using Artificial Intelligence (Machine Learning / Deep Learning), intelligent work-order scheduling, mobile field-force automation, citizen grievance integration, and executive decision-support analytics. By transitioning municipal operations from reactive "break-fix" paradigms to proactive predictive maintenance, the platform minimizes asset downtime, extends physical infrastructure lifespan, reduces operational expenditure, and enhances overall public service quality.

3.3 Definitions

- **Asset Life-Cycle:** The full operational sequence of a municipal physical asset from commissioning, usage, maintenance, retrofitting, to decommissioning.
- **Predictive Maintenance (PdM):** A proactive maintenance strategy that uses data analysis tools and IoT telemetry to detect anomalies in operation and defect indicators before catastrophic failure occurs.
- **Geographic Information System (GIS):** A spatial system designed to capture, store, manipulate, analyze, manage, and present spatial or geographic data.
- **Telemetry:** The automatic recording and transmission of data from remote or inaccessible sources via sensor nodes to an IT network.
- **Work Order:** A formal digital authorization issued to maintenance technicians detailing repair tasks, required inventory, spatial location, and SLA timeframes.

3.4 Acronyms

Acronym	Expansion / Meaning
SRS	Software Requirement Specification
IEEE	Institute of Electrical and Electronics Engineers
GIS	Geographic Information System
IoT	Internet of Things
AI / ML	Artificial Intelligence / Machine Learning
API	Application Programming Interface
SLA	Service Level Agreement
RBAC	Role-Based Access Control
MQTT	Message Queuing Telemetry Transport
RUL	Remaining Useful Life
GDPR	General Data Protection Regulation

3.5 References

- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 12207:2017 Systems and Software Engineering — Software life cycle processes.
- Open Geospatial Consortium (OGC) Standard Specifications for Web Feature Service (WFS) and Web Map Service (WMS).
- ISO 55000:2014 Asset Management — Overview, principles and terminology.

3.6 Overview

The remainder of this document presents the Problem Description (Section 4), Business Domain Analysis (Section 5), Existing System Evaluation (Section 6), Proposed Architecture (Section 7), Stakeholder Matrix (Section 8), Functional & Non-Functional Requirements (Sections 9-10), Use Case Descriptions (Section 11), Data Entities (Section 12), External Interfaces (Section 13), System Features (Section 14), Constraints and Validation (Sections 15-18), Conclusion (Section 19), and complete Planful Unified Modeling Language diagrams (Section 20).

4. Problem Description

Modern municipal corporations govern vast, geographically dispersed physical assets worth billions of dollars. However, municipal administration faces severe systemic inefficiencies caused by legacy operational paradigms. Public assets such as roadways, stormwater drains, street lighting grids, water pipelines, and bridges suffer premature degradation due to delayed maintenance, fragmented paper-based tracking, and complete lack of real-time visibility.

Specifically, conventional municipal management suffers from the following critical failure points:

- **Reactive Maintenance Loops:** Repairs are typically initiated only after a catastrophic asset failure occurs or when citizens register intense complaints, leading to elevated repair costs and public hazards.
- **Lack of Real-Time Infrastructure Health Data:** Decision-makers rely on periodic manual physical audits, which are error-prone, subjective, slow, and labor-intensive.
- **Fragmented Information Silos:** Data regarding asset age, past maintenance, geographic boundary, and operational load is segregated across disconnected departmental files, preventing unified strategic planning.
- **Sub-Optimal Field Force Allocation:** Maintenance work orders are assigned manually without optimization for geographic proximity, skill level, or urgency, causing prolonged resolution turnaround times.
- **Lack of Accountability and Citizen Transparency:** Citizens have limited visibility into civic grievance resolution, leading to public dissatisfaction and diminished trust in municipal authorities.

5. Business Domain Analysis

Municipal Asset Management operates at the intersection of public administration, civil engineering, urban planning, and IoT software architecture. The business domain requires managing heterogeneous asset classes across diverse operational domains:

Asset Category	Examples	Primary Telemetry / Monitoring Vector
Transportation Infrastructure	Asphalt roads, concrete highways, bridges, flyovers, traffic signals	Surface vibration sensors, acoustic strain gauges, visual crack image feeds, traffic load counts.
Utility & Hydrological	Drainage channels, sewage pipes, water distribution mains, pumps	Flow rate meters, water pressure transducers, chemical pH sensors, ultrasonic water level sensors.
Electrical & Energy	LED streetlights, power transformers, municipal solar arrays	Current/voltage telemetry, ambient light sensors, operational temperature, power factor.
Civic & Environmental	Public parks, municipal buildings, recreation grounds, waste bins	Fill-level ultrasonic sensors, ambient air quality indexes, structural inclination sensors.

The primary business objective is to maximize the ROI of public capital investments by extending the functional lifespan of assets, optimizing maintenance operational expenditure (OPEX), and upholding municipal service quality standards (SLAs).

6. Existing System

The existing municipal operational model relies heavily on manual physical inspections, paper registry logs, and reactive citizen complaint logbooks.

6.1 Advantages of Existing System

- **Low initial technology infrastructure and capital expenditure** (No IoT or AI software cost).
- **Simple operational workflow** familiar to legacy staff without requiring digital upskilling.

6.2 Disadvantages of Existing System

- **Extreme Delay in Failure Detection:** Defects remain unnoticed until complete failure occurs.
- **High Long-Term Operational Expenditure:** Emergency repairs cost 3x to 5x more than scheduled preventive interventions.
- **Human Error & Subjectivity:** Manual inspection reports depend heavily on individual surveyor bias and lack standardized quantitative metrics.
- **Zero Predictive Capabilities:** Inability to forecast degradation trends or plan multi-year capital budget allocation based on scientific data.
- **Inconsistent SLA Compliance:** Absence of real-time tracking leads to unverified work order closures by contractors.

7. Proposed System

The proposed **AI-Powered Smart Municipal Asset Management Platform** transitions municipal operations into an automated, data-driven enterprise architecture. It unifies IoT telemetry stream processing, AI predictive modeling, GIS spatial engine, automated work order routing, and citizen engagement.

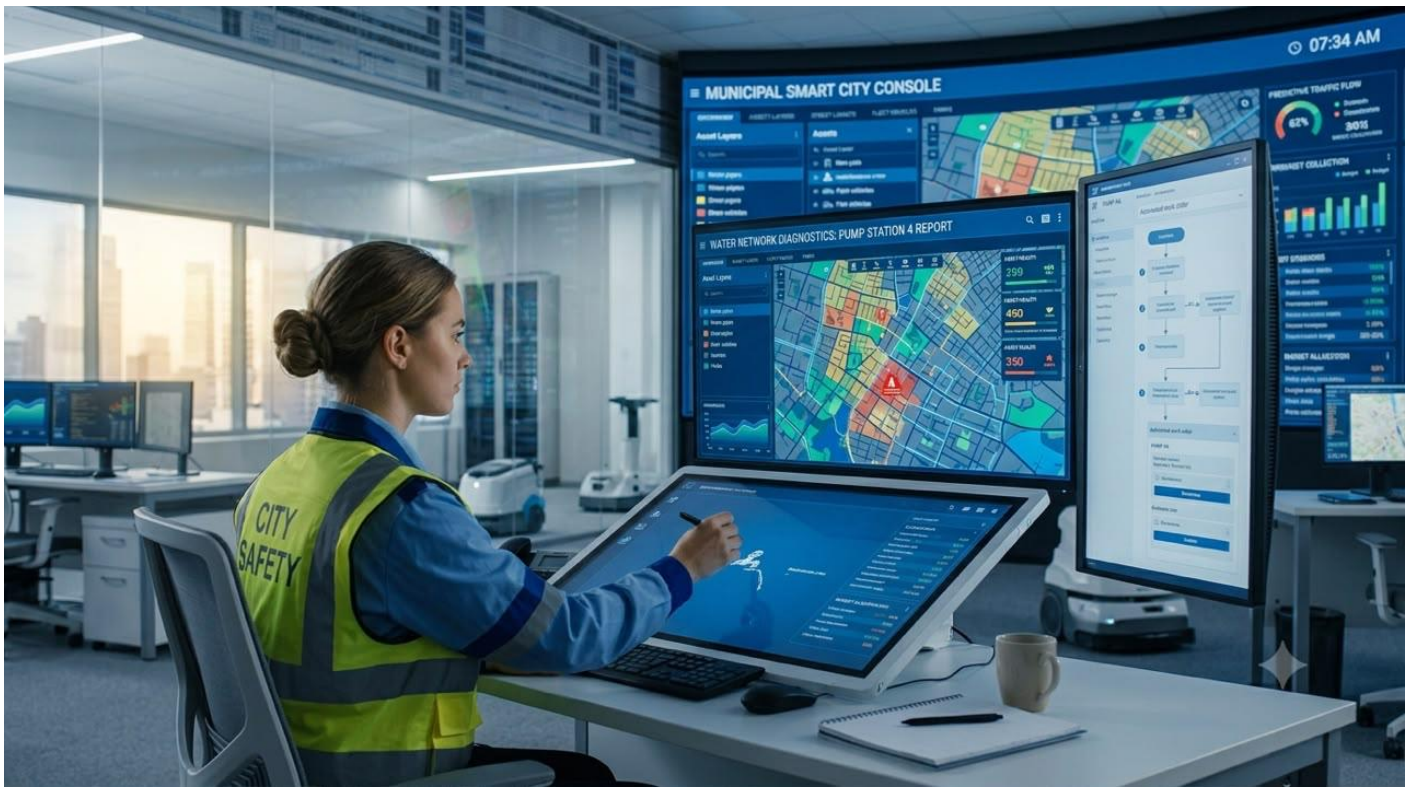
7.1 Advantages of Proposed System

- **Continuous Automated Monitoring:** IoT sensors stream asset telemetry 24/7, enabling instant anomaly detection.
- **AI Predictive Analytics:** ML models compute Remaining Useful Life (RUL) and schedule repairs prior to structural breakdown.
- **Geospatial Intelligence:** GIS visualization maps asset health spatially, enabling optimal route planning for maintenance crews.
- **Cost Optimization:** Minimizes emergency downtime and extends asset service life by up to 35%.
- **Transparency & Citizen Empowerment:** Mobile citizen app allows real-time issue reporting with geo-tagged images and live resolution tracking.

7.2 System Objectives

1. Digitize 100% of municipal asset records into a unified spatial-relational database.
2. Monitor environmental and operational asset conditions continuously via IoT telemetry.
3. Predict asset degradation and calculate Remaining Useful Life (RUL) using AI algorithms.
4. Prioritize work orders dynamically based on critical severity indexes and spatial proximity.
5. Provide interactive multi-layered GIS maps for urban planners and maintenance teams.
6. Automate executive reporting and operational analytics generation.





8. Stakeholders & Roles

Stakeholder	Type	Primary Responsibilities & Key Objectives
Municipal Administrator	Primary Human Actor	Oversees high-level operations, approves capital budgets, monitors departmental SLAs, reviews executive analytics, and configures municipal governance policies.
Maintenance Engineer	Primary Human Actor	Reviews AI-generated predictive maintenance schedules, creates and approves work orders, assigns personnel, and oversees field execution quality.
Field Technician	Primary Human Actor	Executes field repairs, receives mobile work orders, updates task status, captures post-repair photos, and logs material utilization in real time.
Citizens	External Human Actor	Reports infrastructure defects via mobile app, uploads geo-tagged images, tracks complaint progress, and submits satisfaction feedback.
GIS Administrator	Primary Human Actor	Manages spatial layers, updates geospatial assets, verifies coordinate mapping, and maintains OGC GIS service integrations.
AI Prediction Subsystem	System Actor	Ingests continuous telemetry, executes ML degradation algorithms, calculates asset failure probabilities, and auto-generates proactive alerts.
IoT Sensor Fleet	System Actor	Transmits real-time physical telemetry (vibration, temperature, water level, power consumption) over secure MQTT/CoAP networks.
Government Authority	External Governance	Audits municipal performance metrics, reviews compliance reports, ensures alignment with environmental regulations and public funding norms.

9. Functional Requirements

The system must satisfy the following 20 comprehensive functional requirements (FR-01 to FR-20):

Req ID	Requirement Name	Detailed Functional Specification
FR-01	Digital Asset Inventory Management	The system shall allow administrators to create, update, decommission, and query digital records of municipal assets including physical attributes, installation date, warranty status, manufacturer, and initial valuation.
FR-02	IoT Telemetry Data Ingestion	The system shall ingest continuous stream telemetry from deployed IoT sensors using MQTT/HTTPS protocols at scalable frequencies (up to 10,000 events/sec) and store raw data in a time-series database.
FR-03	AI-Driven Health Score Calculation	The system shall execute machine learning models every 6 hours to analyze historical telemetry and compute a dynamic Health Score (0-100) for each registered asset.
FR-04	Predictive Failure Alerting	The system shall trigger automated predictive alerts to Maintenance Engineers when an asset's computed Remaining Useful Life (RUL) falls below specified operational thresholds.
FR-05	Interactive GIS Asset Visualization	The system shall render multi-layered interactive spatial maps (using OpenLayers/Leaflet) displaying asset spatial distribution, real-time health color-coding, and geo-fenced district boundaries.
FR-06	Automated Work Order Generation	The system shall automatically generate draft work orders when an AI predictive threshold or critical IoT anomaly alert is triggered.
FR-07	Work Order Scheduling & Assignment	The system shall provide an automated and manual dispatch engine to route work orders to field technicians based on geographic proximity, skill specialization, and resource availability.
FR-08	Citizen Complaint Lodgement	The system shall provide a public mobile/web interface allowing citizens to submit complaints containing GPS coordinates, categorization, description, and attached camera photos.
FR-09	AI Grievance Categorization	The system shall utilize Natural Language Processing (NLP) and Computer Vision to automatically categorize citizen reports and cross-reference them with existing asset IDs.
FR-10	Mobile Technician Work Order Management	The system shall provide an offline-capable mobile app for field technicians to view assigned work orders, navigate via GPS, log maintenance activity, and capture completion photos.
FR-11	Real-Time SLA Tracking	The system shall track resolution timelines against predefined Service Level Agreements (SLAs) based on priority level and automatically escalate breached tickets to senior management.
FR-12	Role-Based Access Control (RBAC)	The system shall enforce strict granular RBAC restricting user access to data and functional modules based on defined system roles and district jurisdictions.

FR-13	Asset Lifecycle Cost Analysis	The system shall calculate cumulative maintenance costs against original capital cost to recommend whether an asset should be repaired, retrofitted, or completely replaced.
FR-14	Sensor Calibration & Diagnostic Monitoring	The system shall perform automated diagnostic health checks on IoT sensors to detect dead batteries, signal loss, or sensor calibration drift, generating maintenance tickets for sensor hardware.
FR-15	Automated Executive Report Generation	The system shall generate exportable operational, financial, and regulatory compliance reports in PDF, CSV, and XLSX formats on daily, weekly, monthly, and annual schedules.
FR-16	Multi-Channel Notification Dispatcher	The system shall dispatch urgent system alerts and work order updates via SMS, Push Notifications, and Email using integrated third-party gateways.
FR-17	Citizen Feedback and Rating System	The system shall allow citizens to review, rate (1-5 stars), and provide comments on resolved complaints, feeding into municipal quality KPI reports.
FR-18	GIS Spatial Search and Querying	The system shall support spatial queries allowing users to search for assets within a specific radius, polygon district, or along a specific roadway segment.
FR-19	Spare Parts Inventory Integration	The system shall track spare parts stock levels and automatically reserve required parts upon work order creation, alerting storekeepers when reorder thresholds are reached.
FR-20	Audit Logging & Security Archiving	The system shall record an immutable, time-stamped audit log of all system transactions, status changes, manual overrides, and user authentication events.

10. Non-Functional Requirements

10.1 Performance Requirements

- The system dashboard shall load and render GIS layers within less than 2.0 seconds under standard operational network conditions.
- The IoT ingestion pipeline shall process at least 10,000 incoming telemetry events per second with a latency of less than 500ms.
- Database query execution for standard asset search queries shall be completed in under 200 milliseconds.

10.2 Security Requirements

- All data in transit shall be encrypted using TLS 1.3 encryption protocols.
- All sensitive data at rest (user credentials, database storage) shall be encrypted using AES-256 standards.
- The system shall support multi-factor authentication (MFA) for administrative and engineering accounts.

10.3 Reliability Requirements

- The system shall achieve a Mean Time Between Failures (MTBF) exceeding 720 hours of continuous operation.
- The system shall maintain data integrity with zero data loss during sudden server power loss via write-ahead logging (WAL).

10.4 Availability Requirements

- The platform shall ensure 99.9% uptime (24/7/365 operational availability), allowing no more than 8.76 hours of unplanned downtime per year.

10.5 Scalability Requirements

- The cloud-native architecture shall horizontally auto-scale to support up to 1,000,000 registered IoT sensors and 50,000 concurrent active users without degradation of response metrics.

10.6 Maintainability Requirements

- The platform software architecture shall be structured as modular microservices, allowing independent deployment, updating, and maintenance of individual services without full system downtime.

10.7 Usability Requirements

- The web and mobile interfaces shall adhere to Web Content Accessibility Guidelines (WCAG 2.1 Level AA) standards.
- Field technician mobile UI shall be optimized for outdoor readability with high-contrast modes and touch targets > 48px.

10.8 Portability Requirements

- The mobile application shall run natively on Android (v9.0+) and iOS (v14.0+) operating systems.
- The web application shall be cross-browser compatible across Chrome, Firefox, Safari, and Edge.

10.9 Privacy Requirements

- The system shall comply strictly with GDPR and local municipal data privacy standards, anonymizing citizen personal identification data in public analytics.

10.10 Backup & Disaster Recovery Requirements

- Automated full database backups shall occur daily at 02:00 UTC, with continuous incremental write-ahead logging.
- The Recovery Point Objective (RPO) shall be less than 5 minutes, and the Recovery Time Objective (RTO) shall be less than 1 hour in catastrophic failover scenarios.

11. Actors and Use Cases

11.1 Actor Specification

- **Primary Human Actors:** Municipal Administrator, Maintenance Engineer, Field Technician, GIS Administrator, Citizen.
- **Secondary / External Human Actors:** Government Auditor, Inventory Storekeeper.
- **System / Hardware Actors:** AI Prediction Subsystem, IoT Sensor Fleet, Notification Gateway.

11.2 Detailed Use Case Specifications (15 Core Use Cases)

UC-01: Asset Digital Onboarding & Inventory Registration	
Primary Actor	Municipal Administrator / GIS Administrator
Description	Registers a new physical municipal asset into the platform with technical specifications, asset class, spatial coordinates, and vendor details.
Preconditions	User is authenticated with administrative privileges. GIS base map service is operational.
Main Flow	1. Administrator navigates to Asset Management portal and selects 'Register New Asset'. 2. Inputs metadata (Name, Category, Model, Install Date, Expected Lifespan, District). 3. Selects exact geographic coordinate on interactive GIS map. 4. Attaches digital documentation/warranty PDFs. 5. Submits form; system verifies spatial integrity, generates unique Asset ID, and persists record.
Alternative Flow	3a. Invalid GIS coordinates: System prompts administrator to re-pick point or enter valid latitude/longitude manually.
Postconditions	New asset record is created, indexed in spatial database, and visible on GIS dashboard.

UC-02: Real-Time Telemetry Stream Processing	
Primary Actor	IoT Sensor Fleet (System Actor)
Description	Ingests continuous physical measurements (vibration, temperature, water level, power) from IoT hardware installed on assets.
Preconditions	The IoT device is provisioned, authenticated via a TLS certificate, and connected to MQTT Broker.
Main Flow	1. IoT Sensor collects physical sensor readings. 2. Formats payload as JSON with sensor ID, timestamp, and data metrics. 3. Publishes message over encrypted MQTT channel. 4. IoT Gateway ingests, validates schema, enriches with spatial asset ID, and writes to Time-Series DB.
Alternative Flow	4a. Corrupted payload/auth failure: Gateway rejects message, increments security failure metric, and logs anomaly.
Postconditions	Telemetry stored in a time-series database and forwarded to real-time stream engines.

UC-03: AI Maintenance Need & RUL Prediction	
Primary Actor	AI Prediction Subsystem (System Actor)
Description	Executes predictive ML algorithms on historical and real-time telemetry to project Remaining Useful Life (RUL) and structural degradation risk.
Preconditions	At least 72 hours of telemetry data are available for the target asset class.
Main Flow	1. AI Engine fetches rolling window telemetry and operational history. 2. Runs trained ML regression / anomaly detection algorithm. 3. Calculates updated Health Score (0-100) and estimated days of Remaining Useful Life. 4. Stores prediction record in AI Prediction table. 5. If Health Score < 40, triggers automated maintenance alert workflow.
Alternative Flow	2a. Insufficient data points: AI algorithm flags asset status as 'Calibration Required' and requests manual physical check.
Postconditions	Asset Health Score updated; high-risk assets queued for automated work order generation.

UC-04: Automated Predictive Work Order Creation	
Primary Actor	Maintenance Engineer
Description	Transforms AI predictive alerts or critical IoT threshold breaches into structured maintenance work orders.
Preconditions	AI alert generated or manual engineer override initiated.
Main Flow	1. System flags high-priority predictive risk on Engineer Dashboard. 2. Maintenance Engineer reviews AI diagnostics and telemetry trend graphs. 3. Engineer approves automated recommendations or adjusts priority level. 4. System auto-fills work order details, safety protocols, and required spare parts. 5. Work order generated in 'Pending Dispatch' state.
Alternative Flow	3a. Engineer rejects alert: System records rejection reason (e.g., false positive), updating AI training feedback loop.
Postconditions	Work order record created and assigned to operational queue.

UC-05: Dynamic Work Order Dispatch and Technician Assignment	
Primary Actor	Maintenance Engineer
Description	Assigns work order to field technicians optimized by geographic location, current workload, and skill certification.
Preconditions	Work order exists in 'Pending Dispatch' state. Field technicians active on mobile platforms.
Main Flow	1. System queries active field technicians' GPS locations and current task loads. 2. Recommends top 3 optimal technicians based on proximity and skill match. 3. The engineer confirms the assignment. 4. System pushes notification to selected Field Technician's mobile device.

Alternative Flow	4a. Technician declines due to emergency: System re-routes task to next optimal technician.
Postconditions	Work order status updated to 'Assigned'; push notification sent.

UC-06: Citizen Civic Defect Grievance Reporting

Primary Actor	Citizen
Description	Citizen submits public infrastructure flaw report (e.g., pothole, broken streetlight) using mobile app.
Preconditions	Citizen app installed or web portal accessed. Device GPS enabled.
Main Flow	1. Citizen opens app and clicks 'Report Issue'. 2. App auto-captures current GPS coordinates. 3. Citizen selects issue category (e.g., Road, Light, Water), takes photo, and adds brief text. 4. Submits report. 5. System generates unique Tracking Ticket ID and confirms submission.
Alternative Flow	2a. GPS unavailable: Citizen manually drops pin on embedded map interface.
Postconditions	Grievance registered in database and queued for AI computer vision verification.

UC-07: Mobile Field Repair Execution & Closure

Primary Actor	Field Technician
Description	Technician views job on mobile, navigates to location, completes physical repair, logs materials, and submits post-repair proof.
Preconditions	Technician logged into Mobile App; Work order assigned to technician.
Main Flow	1. Technician views job details and clicks 'Start Route' for GPS navigation. 2. Upon arrival, clicks 'Start Repair' (updates status in real-time). 3. Executes physical repair work. 4. Takes post-repair photo, scans barcodes of consumed spare parts, and collects digital signature. 5. Clicks 'Complete Work Order'.
Alternative Flow	4a. Offline mode: App saves data locally and auto-syncs when cellular connectivity restores.
Postconditions	Work order marked 'Pending Verification'; inventory deducted; supervisor notified.

UC-08: Interactive GIS Map Layer Filtering and Spatial Query

Primary Actor	GIS Administrator / Maintenance Engineer
Description	Filters spatial map views by asset category, health index, operational status, or municipal ward boundary.
Preconditions	User authenticated and logged into administrative portal dashboard.

Main Flow	1. User opens GIS Spatial Portal. 2. Selects layer checkboxes (e.g., Water Mains, Streetlights, Critical Roads). 3. Applies filter slider (e.g., Health Score < 50). 4. GIS engine queries database and dynamically updates map layer overlays in real time.
Alternative Flow	4a. Query yields > 50,000 entities: Engine automatically clusters markers for optimal rendering.
Postconditions	Filtered asset spatial view displayed with interactive clickable cluster pins.

UC-09: Executive Dashboard Analytics & Reporting	
Primary Actor	Municipal Administrator
Description	Generates high-level aggregated operational reports, budget utilization charts, and municipal SLA performance trends.
Preconditions	Administrator authenticated; historical maintenance logs present.
Main Flow	1. Administrator accesses Executive Analytics panel. 2. Selects date range, district scope, and asset domain. 3. System aggregates metrics (MTTR, SLA Compliance %, Expenditures). 4. Displays interactive charts and provides 'Export PDF / XLSX' button.
Alternative Flow	4a. Scheduled auto-generation: System emails compiled PDF report to leadership every Monday morning.
Postconditions	Report rendered or exported to local file store.

UC-10: Sensor Health & Diagnostic Tracking	
Primary Actor	Maintenance Engineer / IoT System
Description	Monitors operational health of physical IoT sensors (battery voltage, signal strength, heartbeat latency).
Preconditions	IoT fleet actively transmitting heartbeat packets.
Main Flow	1. Diagnostic worker checks sensor heartbeat timestamps and battery voltage. 2. Flags sensors with low battery (< 15%) or zero signal for > 2 hours. 3. Generates low-priority 'Sensor Inspection' work order automatically.
Alternative Flow	2a. Transient network loss: System retries 3 heartbeat cycles before declaring sensor offline.
Postconditions	Sensor diagnostic record updated; hardware maintenance scheduled.

UC-11: Work Order Quality Verification & Sign-off	
Primary Actor	Maintenance Engineer

Description	Engineer inspects completed field work, verifies uploaded post-repair photos, and officially closes work order.
Preconditions	Work Order status is 'Pending Verification'.
Main Flow	1. Engineer opens pending verification queue. 2. Reviews technician notes, material consumption, and post-repair imagery. 3. Approves work order completion. 4. System marks Work Order as 'CLOSED' and updates Asset Health Score to 100%.
Alternative Flow	3a. Inspection failed: Engineer rejects submission and re-opens work order with corrective instructions.
Postconditions	Work order finalized; asset record status updated; citizen notified if linked to grievance.

UC-12: Citizen Feedback & Rating Submission

Primary Actor	Citizen
Description	Citizen evaluates resolved infrastructure issue, providing 1-5 star feedback and optional comments.
Preconditions	Linked citizen grievance ticket status updated to 'CLOSED'.
Main Flow	1. Citizen receives push notification that grievance is resolved. 2. Opens app, views resolution photo. 3. Selects rating (1 to 5 stars) and enters review text. 4. Submits feedback; system archives response.
Alternative Flow	3a. Rating < 2 stars: System automatically alerts Citizen Relationship Manager for follow-up review.
Postconditions	Feedback stored and linked to department KPI analytics.

UC-13: Spare Parts Inventory Reservation & Tracking

Primary Actor	Maintenance Engineer / Inventory System
Description	Automatically reserves required replacement components (e.g., LED drivers, pipe valves) upon work order creation.
Preconditions	Work order created specifying bill of materials.
Main Flow	1. System checks inventory database for requested spare part SKU. 2. Decrements available stock quantity and places parts in 'Reserved' state. 3. If available stock < reorder threshold, alerts storekeeper to issue purchase order.
Alternative Flow	2a. Stock out of stock: System alerts engineer that assignment must be delayed due to material shortage.
Postconditions	Inventory levels updated accurately; storekeeper notified if reorder point reached.

UC-14: Role-Based User Access & Permission Management	
Primary Actor	Municipal Administrator
Description	Configures user accounts, assigns system roles, and defines jurisdictional ward access permissions.
Preconditions	Administrator authenticated with root privilege.
Main Flow	1. Administrator navigates to User Management console. 2. Creates new user account or selects existing profile. 3. Assigns role (e.g., Field Tech, Engineer, GIS Admin) and assigns specific Ward/District scopes. 4. Saves profile; system updates RBAC authorization tables.
Alternative Flow	3a. User revoked: Admin disables account; active sessions terminated instantly across all platforms.
Postconditions	User access rights restricted strictly to assigned role and geographic jurisdiction.

UC-15: Regulatory Performance Audit Log Export	
Primary Actor	Government Authority / Auditor
Description	Generates cryptographically verified operational logs and spending reports for government regulatory audits.
Preconditions	Auditor authenticated with read-only compliance access.
Main Flow	1. Auditor logs into Compliance Portal. 2. Selects fiscal year, project category, and audit scope. 3. Requests audit bundle. 4. System compiles immutable audit logs, digital signatures, work orders, and financial transactions into an encrypted zip archive.
Alternative Flow	4a. Log mismatch detected: System flags security audit exception for administrative investigation.
Postconditions	Exported verified audit trail package generated for regulatory oversight.

12. Data Requirements & Schema Structure

The platform database architecture requires a hybrid relational (PostgreSQL/PostGIS) and time-series (TimescaleDB) database engine. Below are the key entity specifications:

Entity Name	Primary Attributes & Data Types	Relationships & Constraints
Asset	asset_id (UUID, PK), name (VARCHAR), category (ENUM), install_date (DATE), expected_life_yrs (INT), current_health_score (FLOAT), viscid (UUID, FK)	One-to-Many with Sensors, WorkOrders, AI_Predictions. Foreign Key to GIS_Location.
User	user_id (UUID, PK), username (VARCHAR), email (VARCHAR), password_hash (VARCHAR), role (ENUM), ward_jurisdiction (INT)	One-to-Many with WorkOrders (Assigned/Created), Grievances.
Sensor	sensor_id (UUID, PK), asset_id (UUID, FK), sensor_type (VARCHAR), battery_level (FLOAT), firmware_version (VARCHAR), status (ENUM)	Many-to-One with Asset. One-to-Many with Telemetry_Readings.
GIS_Location	gis_id (UUID, PK), latitude (DECIMAL), longitude (DECIMAL), elevation (FLOAT), geo_polygon (GEOMETRY), district_name (VARCHAR)	One-to-One with Asset and Grievance records. PostGIS Spatial Index enabled.
Work_Order	work_order_id (UUID, PK), asset_id (UUID, FK), assigned_tech_id (UUID, FK), created_by (UUID, FK), priority (ENUM), status (ENUM), scheduled_date (TIMESTAMP)	Foreign Keys to Asset, User (Technician), User (Engineer). One-to-Many with Inspection_Reports.
Inspection_Report	report_id (UUID, PK), work_order_id (UUID, FK), tech_id (UUID, FK), remarks (TEXT), before_img_url (VARCHAR), after_img_url (VARCHAR), completed_at (TIMESTAMP)	Many-to-One with Work_Order and User.
AI_Prediction	prediction_id (UUID, PK), asset_id (UUID, FK), predicted_rul_days (INT), failure_probability (FLOAT), model_version (VARCHAR), generated_at (TIMESTAMP)	Many-to-One with Asset. Used for trend analysis and automated alerts.
Maintenance_Request (Grievance)	request_id (UUID, PK), citizen_id (UUID, FK), gis_id (UUID, FK), category (VARCHAR), description (TEXT), status (ENUM), created_at (TIMESTAMP)	Foreign Keys to User (Citizen) and GIS_Location. Can link to Work_Order.

13. External Interface Requirements

13.1 User Interface (UI) Requirements

- **Web Admin Dashboard:** Responsive, dark/light theme capable interface built using modern frontend frameworks (React/Vue). Features split-screen viewing (GIS map on left, analytical cards on right).
- **Mobile Field App:** Lightweight Android/iOS native app with large touch controls, offline sync capabilities, voice-to-text input for technician notes, and camera capture.
- **Citizen Portal:** Simplified 3-step reporting workflow with auto-geolocation and status progress tracking timeline.

13.2 Hardware Interfaces

- **IoT Sensors:** Interface with structural vibration sensors, ultrasonic water sensors, power meters, and temperature probes via RS-485, Modbus, or GPIO pins.
- **Mobile Handhelds:** Interface with smartphone GPS modules, camera sensors, and Bluetooth barcode/NFC scanners for asset tag reading.

13.3 Software Interfaces

- **GIS Engine API:** Integration with OpenStreetMap, Esri ArcGIS, or GeoServer via standard OGC WMS/WFS protocols.
- **SMS/Push Gateway:** RESTful API interfaces with Twilio and Firebase Cloud Messaging (FCM) for multi-channel alerts.
- **ERP / Financial System API:** SAP/Oracle ERP REST integration for synchronization of inventory purchase requisitions and municipal capital budgeting.

13.4 Communication Interfaces

- **MQTT Protocol:** Lightweight messaging for low-bandwidth, high-latency IoT sensor communications over cellular LTE-M / NB-IoT / LoRaWAN networks.
- **HTTPS / REST / WebSockets:** Web-to-backend communication encrypted with TLS 1.3 for real-time dashboard updates and mobile API transactions.

14. Detailed System Features

14.1 AI Prediction Engine

The AI engine utilizes Random Forest Regression, Long Short-Term Memory (LSTM) neural networks, and anomaly detection algorithms to model physical asset degradation. By ingesting historical telemetry, environmental conditions, and age profiles, it computes Remaining Useful Life (RUL) and assigns failure probability scores. High-risk predictions automatically trigger draft work orders.

14.2 Real-Time IoT Monitoring Subsystem

An enterprise streaming architecture (Apache Kafka + TimescaleDB) ingests high-frequency IoT data. The subsystem evaluates telemetry against threshold boundary conditions (e.g., pipe vibration > 4.5g, streetlight current = 0A). Out-of-bounds readings generate real-time critical alerts within 500 milliseconds.

14.3 GIS Mapping & Spatial Analysis Module

Integrated with PostGIS spatial indexes, this module renders interactive map overlays. It supports vector tile rendering, heatmaps of frequent asset failures, buffer zone calculations (e.g., assets within 100 meters of a river), and route optimization for maintenance vehicles.

14.4 Executive & Operational Dashboards

Presents real-time Key Performance Indicators (KPIs) including Mean Time to Repair (MTTR), SLA compliance rate, active work order status counts, and ward-wise budget consumption. Includes interactive drilling down from high-level city views to individual asset sensor streams.

14.5 Mobile Field Asset Tracking & Work Order Engine

Empowers field crews with digital work orders. Supports offline operation where field reports and photos are cached on-device and auto-synchronized when network connection is established. Includes barcode/QR scanning to verify physical asset identification prior to starting work.

14.6 Automated Report Generation Subsystem

Generates customizable operational, financial, and compliance reports. Schedules automated weekly/monthly PDF generation and distribution via email to municipal stakeholders and government audit committees.

14.7 Citizen Complaint Portal

A public-facing mobile/web portal enabling citizens to submit municipal infrastructure issues. Utilizes AI vision models to auto-detect duplicate reports within a 50-meter radius, consolidating citizen tickets under a single work order.

15. System Assumptions

- Municipal field technicians possess standard Android/iOS smartphones equipped with functional cameras and GPS modules.
- Deployed IoT sensor hardware is equipped with persistent power sources or solar/battery cells designed for a minimum 3-year operating lifespan.
- Cellular network coverage (4G/5G/NB-IoT) is available across at least 95% of the municipal ward geographic territory.

16. System Constraints

- **Technical Constraints:** System must operate within cloud-native containerized environments (Kubernetes/Docker) and comply with open OGC spatial standards.
- **Operational Constraints:** Field staff must complete a mandatory 4-hour training module on the mobile platform before access is granted.
- **Economic Constraints:** Total initial cloud infrastructure and software operational costs must remain within the allocated municipal annual software budget limit.
- **Legal Constraints:** System data handling must adhere to local municipal privacy laws and national cyber security frameworks.
- **Time Constraints:** Complete enterprise system deployment, testing, and field rollout must be completed within 12 calendar months from project kickoff.

17. Requirement Validation & Verification

Quality Criteria	Verification Strategy & Methodology
Completeness	Validated against IEEE 830 standards ensuring all structural sections, use cases, entity models, and interfaces are fully articulated.
Consistency	Cross-referenced functional requirements with use case flows and entity schema to eliminate contradictory system behavior definitions.
Unambiguity	Formulated requirements using clear, quantitative 'shall' statements with explicit numeric targets (e.g., response time < 2.0s, 99.9% uptime).
Non-Redundancy	Reviewed and normalized requirements to ensure each functional capability is uniquely defined under a single requirement ID.
Traceability	Established a Requirements Traceability Matrix (RTM) mapping Business Objectives → Functional Requirements → Use Cases → Database Schema.

18. Suggested Future Improvements

- **Drone Inspection Integration:** Incorporate autonomous drone video feeds and Computer Vision photogrammetry for automated structural inspection of high bridges and tall municipal buildings.
- **Augmented Reality (AR) Repair Assistance:** Provide field technicians with AR headsets overlaying underground pipe and cable schematics onto physical terrain in real time.

Conclusion

The AI-Powered Smart Municipal Asset Management and Maintenance Platform specifies an advanced, robust, and scalable software solution designed to solve modern urban infrastructure deterioration. By combining IoT continuous telemetry, AI predictive modeling, high-precision GIS spatial visualization, automated work order routing, and citizen engagement, the platform establishes a modern paradigm for smart civic governance. Implementation of this specification will directly lead to extended public asset lifespans, reduced emergency maintenance costs, optimized field workforce productivity, and significantly enhanced citizen satisfaction.

19. System Architectural Diagrams (PlantUML)

Below are the complete, syntactically valid Plenum specifications for the core visual architectural models required for system engineering and implementation.

19.1 Use Case Diagram

```
@startuml
left to right direction skiagram
package Style rectangle
skinparam actor Style solid

actor "Municipal Admin" as Admin
actor "Maintenance Engineer" as Eng
actor "Field Technician" as Tech
actor "Citizen" as Cit
actor "GIS Admin" as GISAdmin
actor "IoT Sensor Fleet" as IoT
actor "AI Subsystem" as AI

rectangle "Smart Municipal Asset Management Platform" {
    usecase "UC-01: Register Asset" as UC1
    usecase "UC-02: Stream Telemetry" as UC2
    usecase "UC-03: Predict Degradation & RUL" as UC3
    usecase "UC-04: Create Work Order" as UC4
    usecase "UC-05: Dispatch Work Order" as UC5
    usecase "UC-06: Report Civic Issue" as UC6
    usecase "UC-07: Execute Field Repair" as UC7
    usecase "UC-08: Filter GIS Layers" as UC8
    usecase "UC-09: Generate Analytics" as UC9
    usecase "UC-10: Sensor Health Check" as UC10
    usecase "UC-11: Verify & Close Work Order" as UC11
    usecase "UC-12: Submit Feedback" as UC12
}

Admin --> UC1
Admin --> UC9
Admin --> UC8

IoT --> UC2
IoT --> UC10

AI --> UC3
UC3 .-> UC4 : <>

Eng --> UC4
Eng --> UC5
Eng --> UC11

Tech --> UC7

Cit --> UC6
Cit --> UC12

GISAdmin --> UC1
GISAdmin --> UC8
@enduml
```